

EMBEDDED CONTROL, CAN & CHARACTERIZATION REPORT

TI Piccolo-Based Dual-Motor Dynamometer Testbed

PUBLIC ENGINEERING RELEASE v1.4

Document ID	HES-DYNO-SW-001
Prepared by	Herder Elektronische Systemen
System basis	TI C2000 Piccolo F28069M dual-PMSM dynamometer testbed
Purpose	Describe the real-time control architecture, CAN migration, host workflows, data-conditioning method, measured results, and evidence limitations that support the companion FRD.

DOCUMENT POSITIONING

Independent engineering documentation for a platform built with Texas Instruments development hardware. This document is not authored, approved, or sponsored by Texas Instruments. Product names and trademarks remain the property of their owners.

1. Document Relationship

The companion Functional Requirements & Validation Report is the authoritative public baseline for requirements, verification methods, status, and traceability. This report explains how the embedded, communications, host, and characterization evidence supports that baseline.

2. Engineering Outcome

An inherited dual-motor bench with mismatched connection documentation was converted into a repeatable embedded-controls and characterization platform. The work corrected physical phase and encoder interfaces, restored dual-machine FOC operation, stabilized host-target pacing, built fixed-dwell and compressed drive-cycle command generation, implemented a CAN command/response path, and produced reviewable efficiency-map evidence.

DUAL-MOTOR DYNAMOMETER SYSTEM ARCHITECTURE

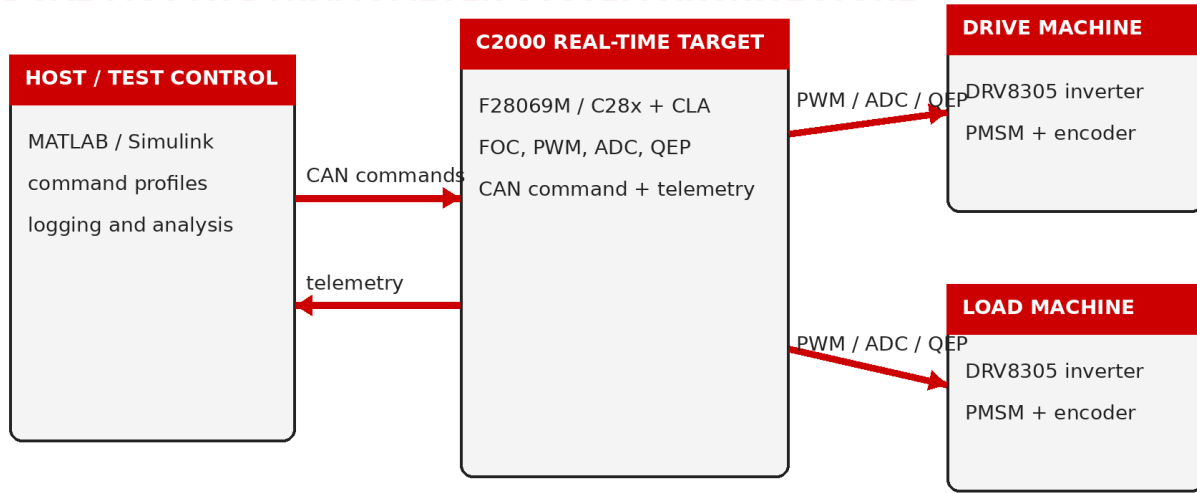


Figure 1. Control and data-flow architecture.

3. Real-Time Target Responsibilities

- Execute the motor-control loop, PWM generation, project-specific ADC-synchronized acquisition, QEP position/speed capture, and command interpretation.
- Maintain separate drive-machine and load-machine control paths.
- Receive packed host commands over CAN and expose response/telemetry values.
- Keep time-critical behavior independent of host plotting and GUI refresh timing.

4. CAN Host-Target Interface

Direction	Identifier	Payload	Public role
Host to target	0x100	4 x 16-bit values	Command transport: speed, load/current, enable/state, and selection/configuration fields.
Target to host	0x101	4 x 16-bit values	Response/echo and selected target values for integration verification.

The retained source evidence shows standard CAN mailbox configuration for these identifiers and byte packing/unpacking for four 16-bit values. Exact private scaling tables and the complete telemetry dictionary are not published.

5. Host Workflows

Workflow	Purpose	Evidence
Fixed-dwell efficiency test	Hold defined speed/load operating points long enough to accumulate valid samples.	efficiency_test_setup.m
Drive-cycle test	Create a bounded, compressed FTP-style speed command and corresponding load command.	drive_cycle_setup.m
Efficiency processing	Crop, align, screen, compute, bin, count, and visualize efficiency data.	motor_efficiency_processing.m
Run export and plotting	Save selected sessions and compare runs.	host-software utilities

6. Characterization Method

Signals captured across selected debug configurations are cropped to a common interval and interpolated onto a common time base. The workflow computes electrical and mechanical power, applies validity masks, bins accepted samples by speed and torque, stores sample counts, and produces measured and interpolated views. Interpolated surfaces are explicitly presentation aids rather than additional measured evidence.

Screening rule	Threshold
Electrical input power	> 10 W
Mechanical output power	> 3 W
Torque	> 0.045 N m
Efficiency	Finite and physically plausible
Speed/torque grid	0-4000 rpm; 0-0.25 N m

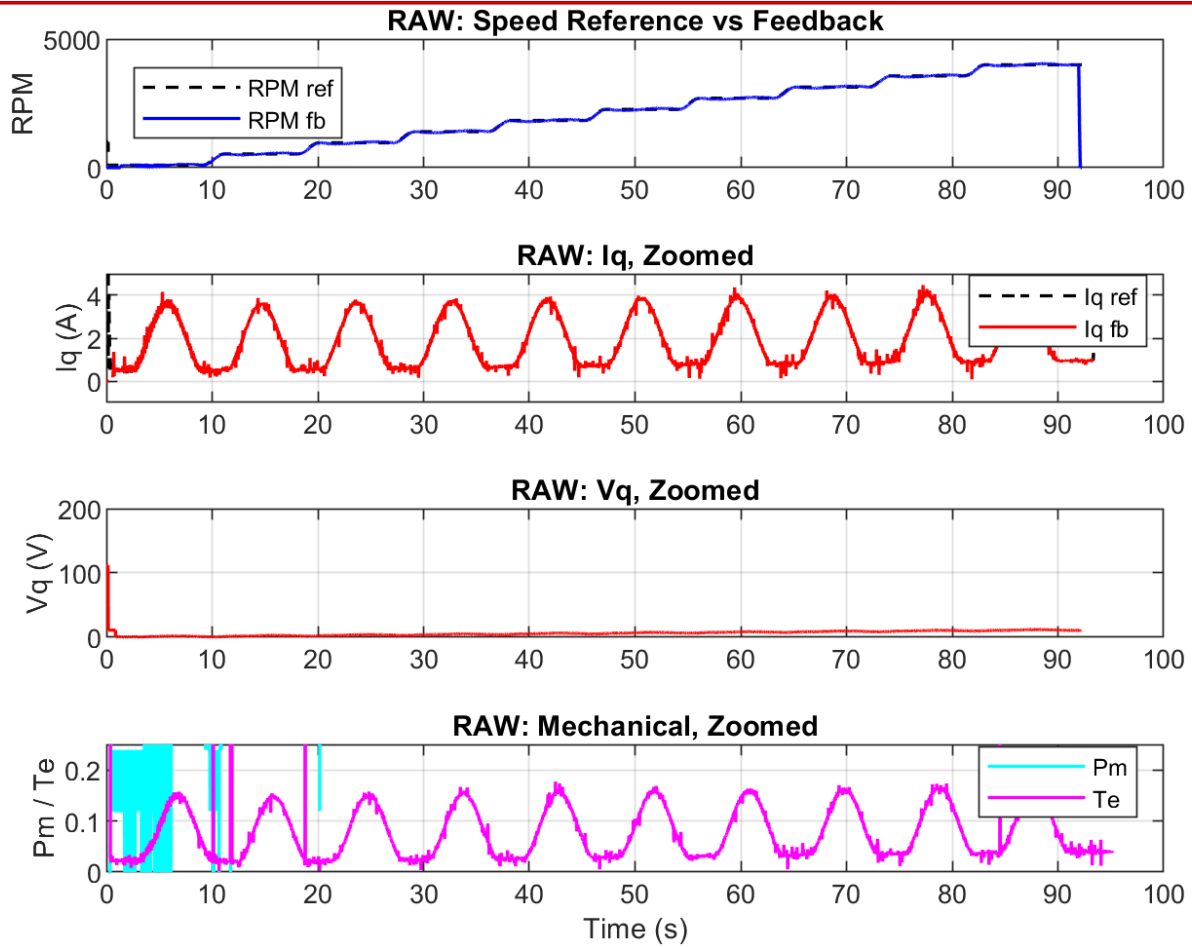


Figure. Raw signal review.

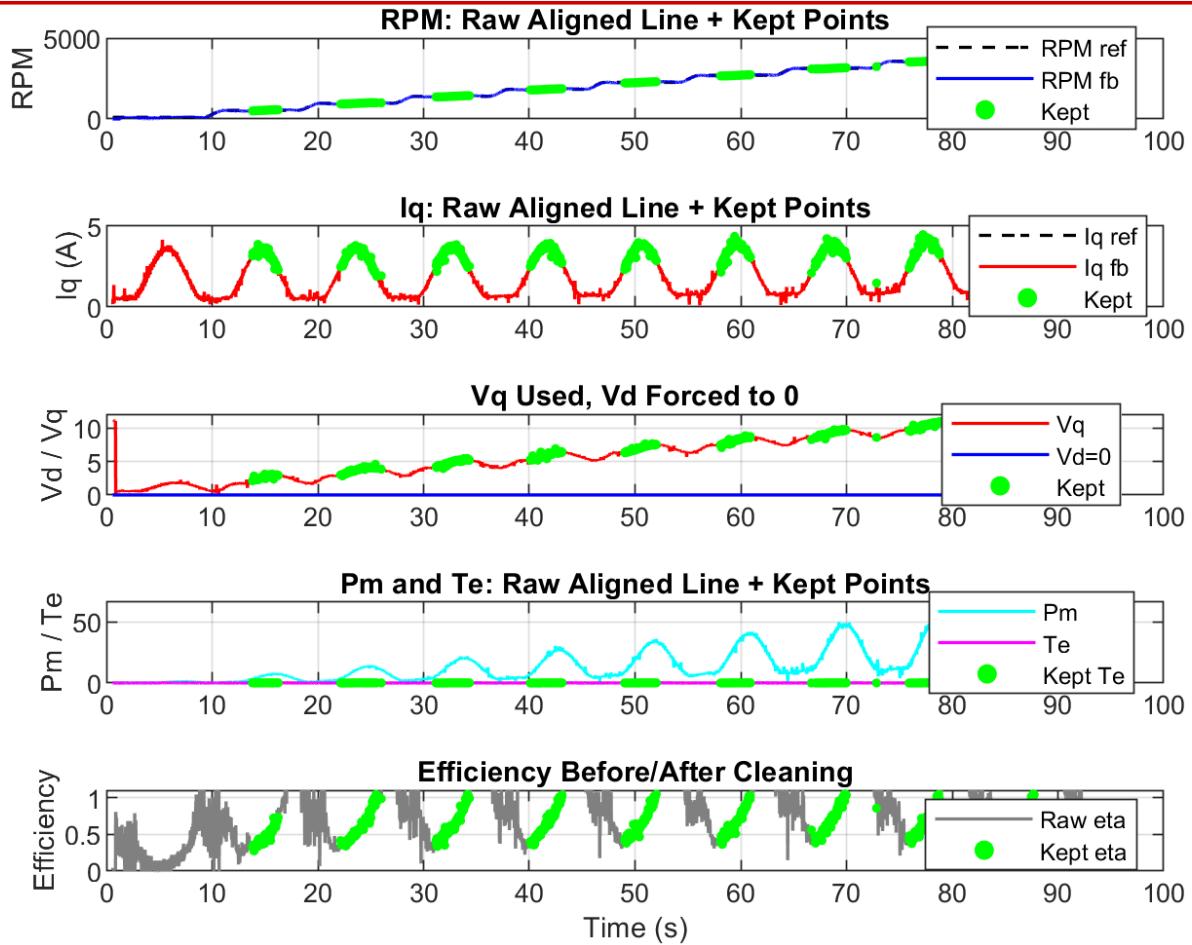


Figure. Cleaning overlay showing retained and rejected regions.

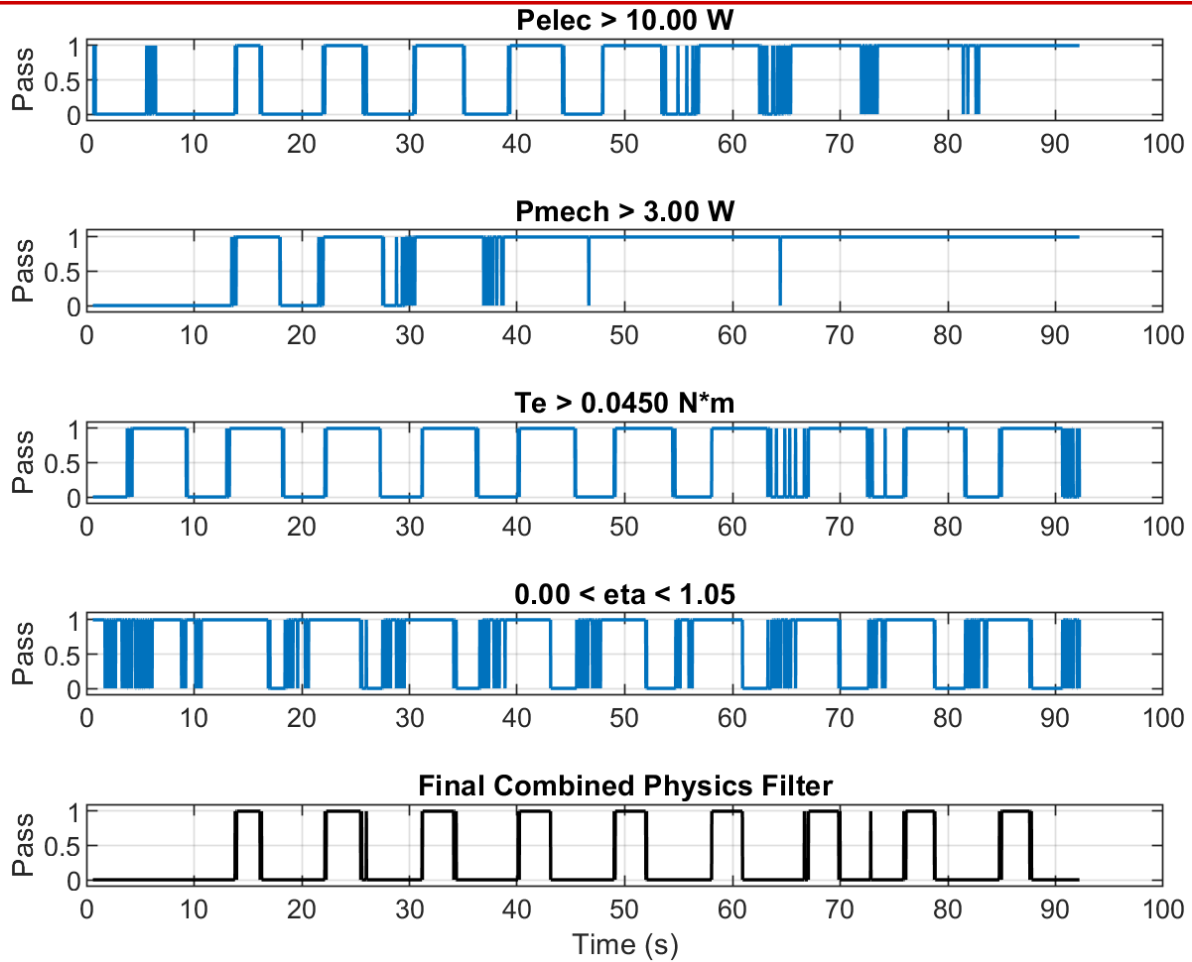


Figure. Validity masks before derived calculations.

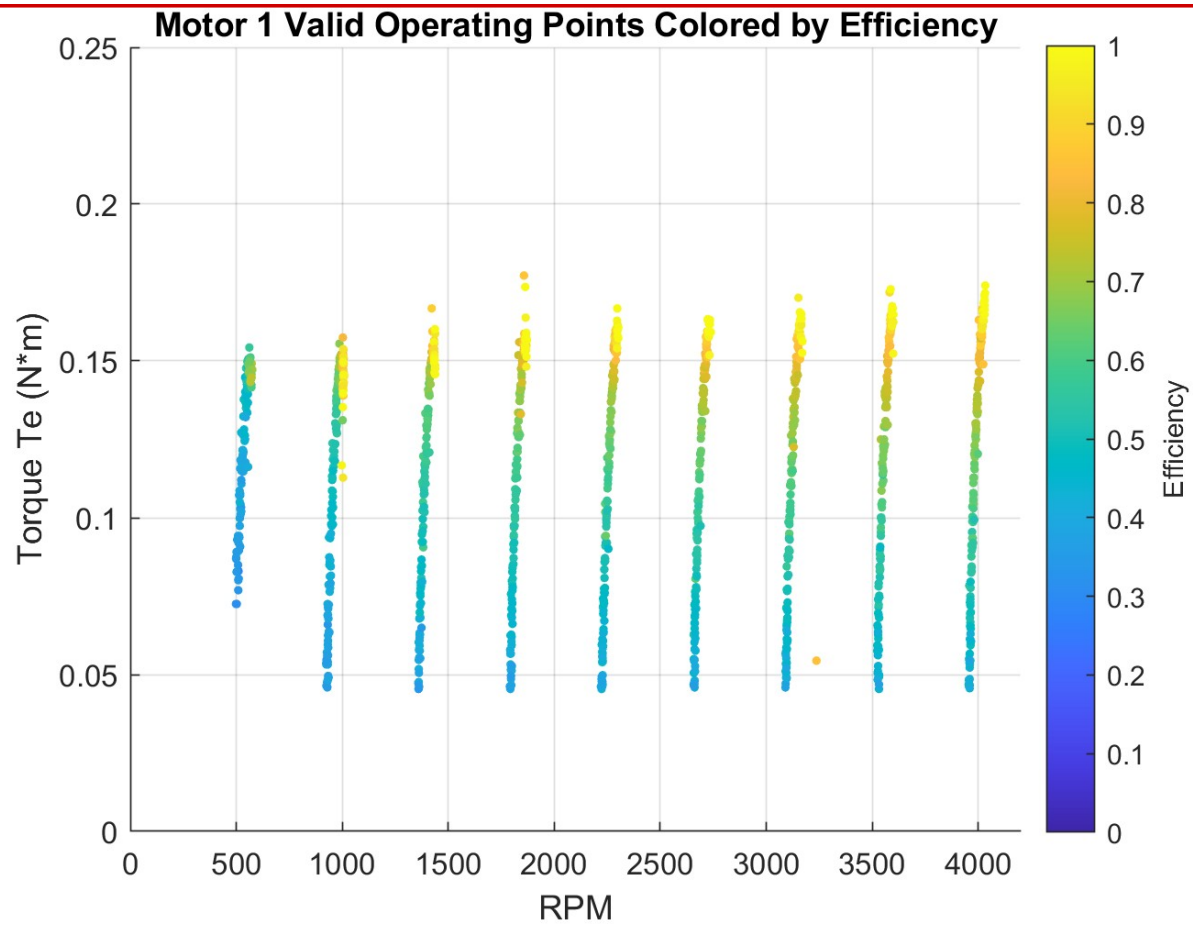


Figure. Accepted measured operating points.

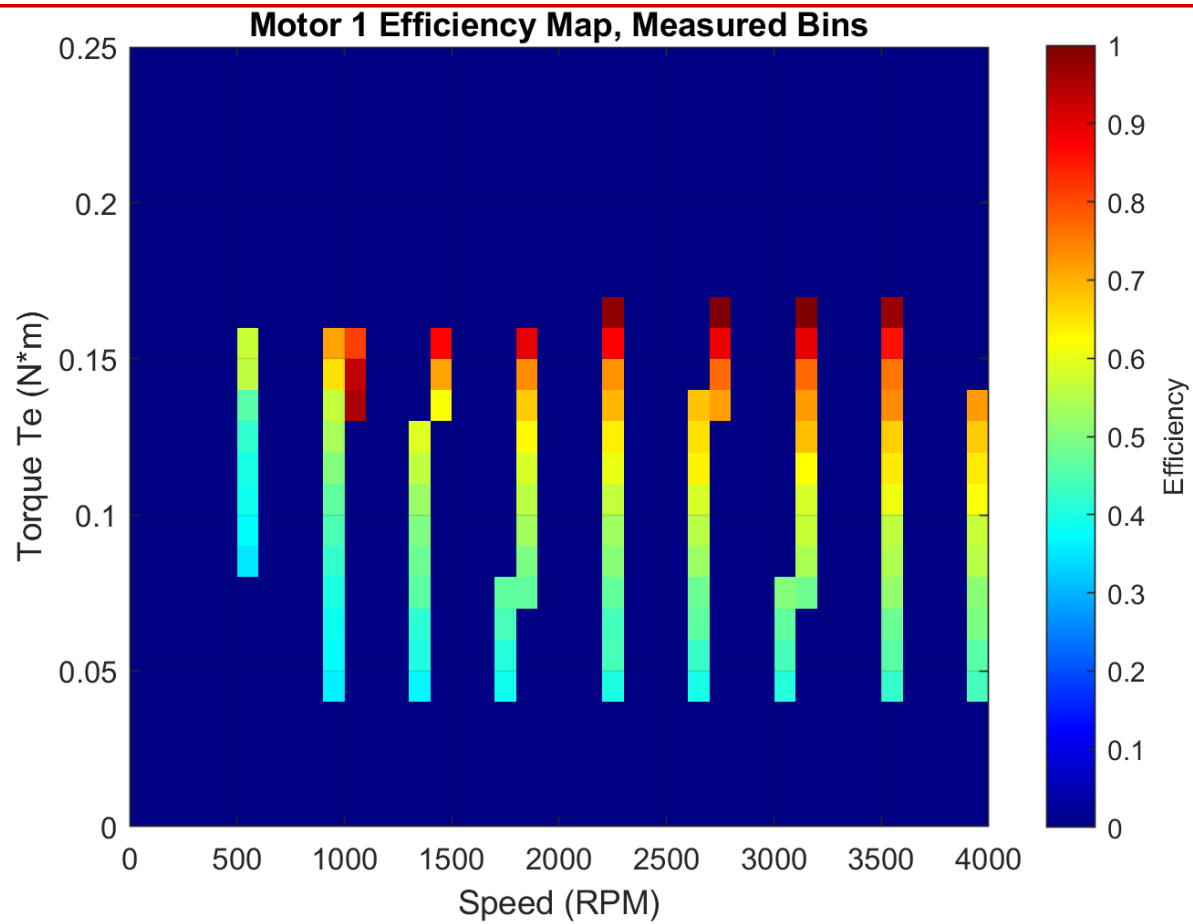


Figure. Measured-bin efficiency map.

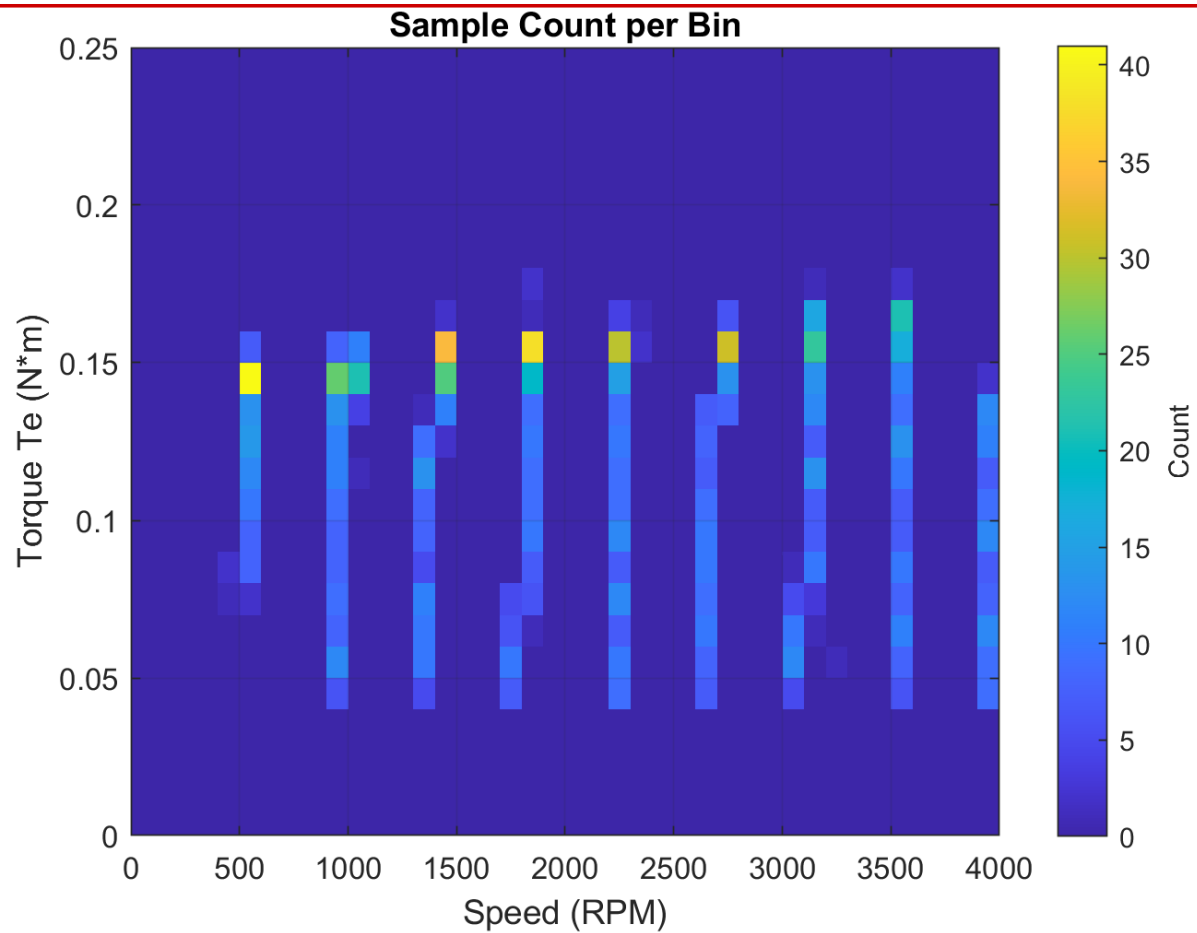


Figure. Evidence density by bin.

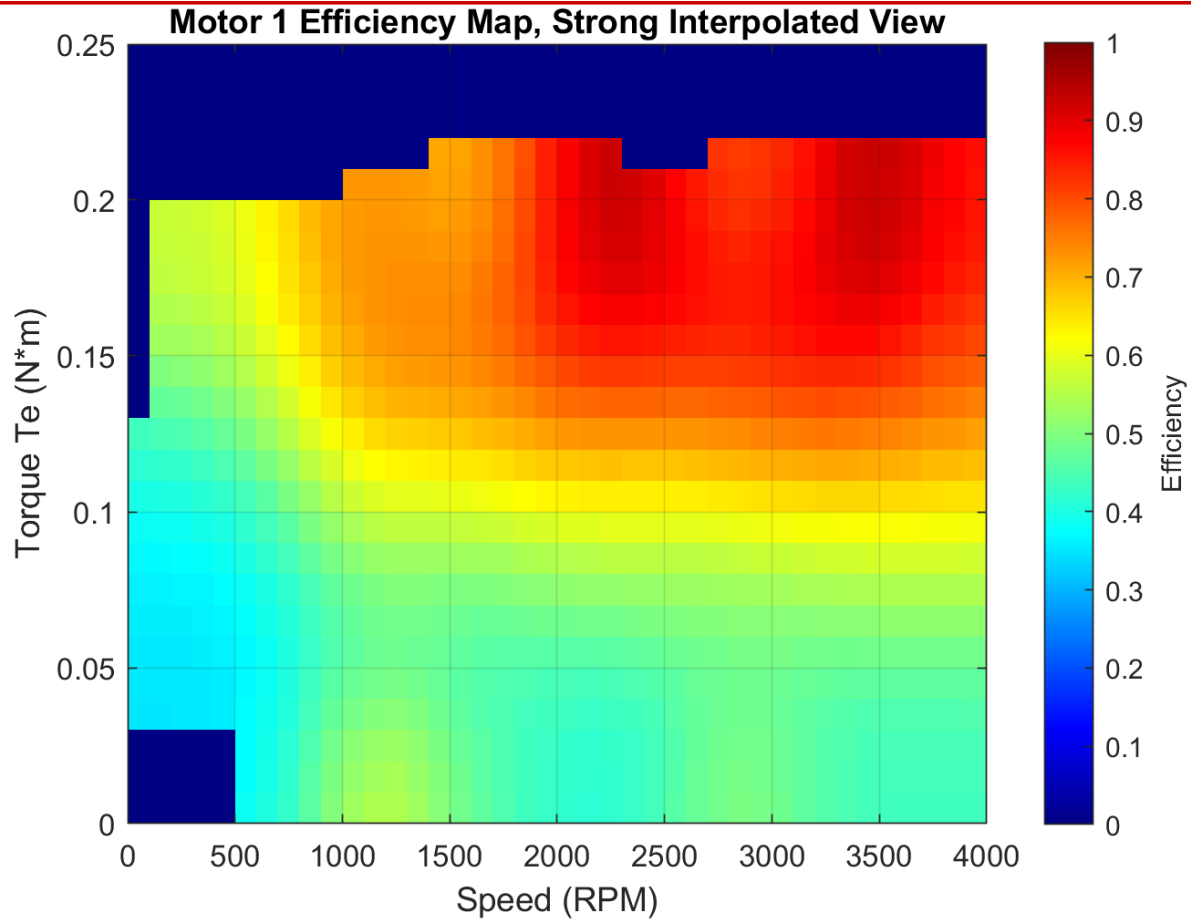


Figure. Interpolated visualization.

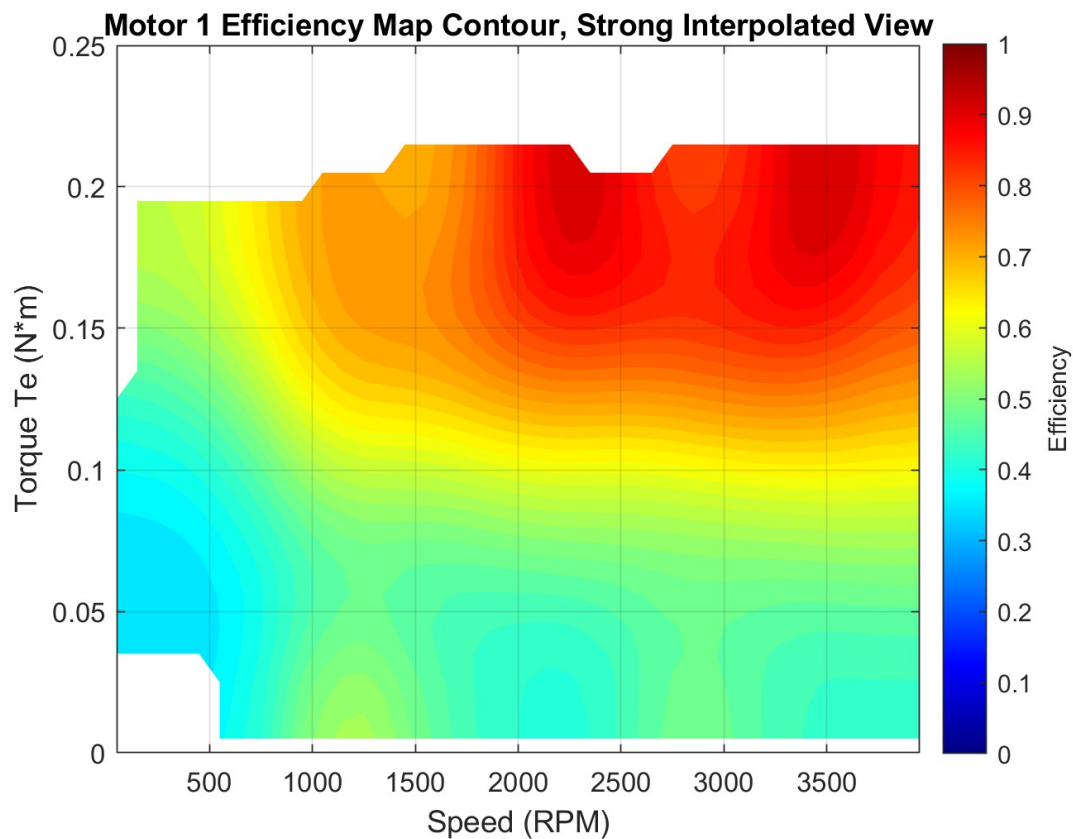


Figure. Contour visualization of the interpolated surface.

7. Demonstrated Outcomes and Interpretation Boundaries

Outcome	Supported statement
Physical recovery	Phase order and encoder/QEP mapping were corrected against the actual platform.
Control bring-up	Dual-machine closed-loop operation was restored on the F28069M target.
Communications	CAN command and response transport was demonstrated with retained mailbox and packing evidence.
Data quality	Non-finite and implausible samples are screened before map generation.
Traceability	Measured bins, sample counts, and interpolated views are retained as distinct evidence products.
Boundary	No production safety, calibrated metrology, EMC, environmental, or functional-safety qualification is claimed.

8. Release Status

This report is released with FRD revision 1.4. The public repository is a capability showcase and engineering evidence package, not a turnkey firmware distribution. Selected sanitized Simulink examples are included; complete target models, binaries, private scaling tables, raw data, and generated implementation trees remain controlled.

Rev	Date	Author	Description
1.1	2026-08-03	HES	Reformatted in TI-inspired application-note style and aligned to the new FRD and traceability model.
1.2	2026-08-03	HES	Added selected sanitized Simulink examples and aligned the companion report to FRD revision 1.4; no substantive technical findings changed.

9. Custom Low-Level ADC Integration

Most application behavior is represented by editable Simulink models, but the target also contains a focused custom C layer for F28069M ADC peripheral configuration. This source is included under embedded-control/custom-firmware/ and is not a renamed generated support file.

9.1 Added acquisition resources

SOC pair	Inputs	Trigger	Completion	Role
SOC0/SOC1	ADCINA0 / ADCINB0	ePWM1	EOC1 -> ADCINT1	Baseline synchronized pair
SOC2/SOC3	ADCINA3 / ADCINB3	ePWM4	EOC3 -> ADCINT2	Baseline synchronized pair
SOC4/SOC5	ADCINA0 / ADCINB0	Software	Polled / model-controlled	Software-triggered pair
SOC6/SOC7	ADCINA3 / ADCINB3	Software	Polled / model-controlled	Software-triggered pair
SOC12/SOC13	ADCINA2 / ADCINB2	ePWM1	EOC13 -> ADCINT5	Added instrumentation pair
SOC14/SOC15	ADCINA1 / ADCINB1	ePWM1	EOC15 -> ADCINT9	Added instrumentation pair

The project-specific additions are SOC12/SOC13 and SOC14/SOC15. Both pairs are synchronized to ePWM1 so the additional instrumentation channels are sampled against target-side timing rather than host execution timing.

9.2 Integration boundary

- The Simulink target remains the primary editable application/control source.
- F28069M_ADC_Custom_Config.c provides the project-specific register-level SOC, trigger, acquisition-window, and ADC interrupt configuration.
- MathWorks-generated ADC support files, TI device-support source, object files, and binaries remain excluded.
- A compatible C2000 support package, device headers, and the referenced target model are required to compile and deploy the complete target.

9.3 Controlled engineering claim

The demonstrated contribution is custom C2000 ADC peripheral integration for two additional simultaneous analog-input pairs, including PWM-triggered sampling and end-of-conversion interrupt routing. The release does not claim a from-scratch ADC driver or ownership of TI/MathWorks support libraries.